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Original article

Determination of Medical Faculty Clinical Students' Attitudes Towards Patient Safety: A Mixed Design Study

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ABSTRACT

Background: The study was planned to determine the attitudes of medical faculty students in clinics toward patient safety.
Methods: It is a mixed-design study using quantitative and qualitative methods. The "student information form" and the "Attitudes toward Patient Safety Culture Questionnaire (APSQ)" created using "Google Forms" were e-mailed to clinical students of the medical faculty. Quantitative data were collected from 125 students. Then, semi-structured focus group interviews were conducted with six students from each class. The data were obtained through video and audio recordings of the 45-minute interviews through the Zoom program.
Results: According to the results, 86% of the 125 students did not receive any training on patient safety. The total score of the students on the APSQ was 94.87 ± 14.32. Six main themes and twenty sub-themes emerged from the qualitative data. The main themes are students' awareness of patient safety, students' misconceptions about "patient safety," students' knowledge about the scope of patient safety, students' experiences with medical errors, deficiencies in educational processes on patient safety, and suggestions for educational approaches to patient safety.
Conclusions: After completing undergraduate medical education, physicians typically assume roles as general practitioners in various healthcare settings, such as hospital's emergency departments, home care, or community health centers. The lack of awareness among medical students about patient safety is a significant concern that needs to be addressed. As stakeholders in their education, students recognize the importance of being well-prepared to provide safe and effective patient care in their future roles as physicians.
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1. Introduction

Patient safety is the avoidance of harm to patients, and it is the basis of the provision of quality healthcare [1]. There is a widespread consensus that quality health care must be effective, safe, and patient-centered [2]. To successfully implement patient safety strategies, it is crucial to have well-defined policies, access to relevant data to drive safety improvements, and a competent team of healthcare professionals [3].

A significant report by the United States Institute of Medicine in 2000 revealed that approximately 98,000 individuals lost their lives annually due to medical errors [4]. Despite the passage of 23 years since the report's issuance, it is concerning to observe that medical errors continue to be documented and linked to fatalities [5–7]. Notably, the World Health Organization highlights that a considerable portion of these medical errors, ranging from 50% to 70.2%, can be prevented [3].

Medical errors predominantly occur due to preventable factors [7], with significant contributors being attributed to communication gaps, leadership issues, and human factors [8]. Ensuring health care is provided in a safe environment is a fundamental patient right, applicable to both healthy individuals and those facing illnesses.

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Addressing and mitigating harm caused by medical errors is a global priority. Medical students play a crucial role in this endeavor by being able to identify unsafe conditions, report errors, and near-misses systematically, investigate main causes, and implement improvements, all while acknowledging the inherent fallibility of human beings. To achieve these objectives, there is a pressing need for patient safety to be integrated into the medical student curriculum [9,10]. The World Health Organization (WHO) and the American Association of Medical Colleges (AAMC) have reported that patient safety and quality improvement issues should be added to the undergraduate medical education curriculum [11,12]. There are limited programs related to patient safety in Turkey. The center where the study was conducted holds a prominent position in addressing the country's physician demand. A limited number of studies have explored the attitudes and opinions of medical faculty students concerning patient safety [8,13,14]. Nevertheless, no study has been conducted explicitly within our institution.

This study was conducted to determine the attitudes of medical faculty clinical year students toward patient safety. Determining students' attitudes toward patient safety will provide basic data for future educational activities.

Research questions

1. What is the level of attitude of medical faculty clinical period students toward patient safety culture?
2. What are the views of medical faculty clinical period students on patient safety?

2. Materials and methods

The study was designed in a mixed design, including descriptive and qualitative research methods in two stages.

During the first stage of the study, data collection was achieved through two main instruments. The "Attitudes toward Patient Safety Culture Questionnaire" (APSQ) and "Student Descriptive Information Form," including sociodemographic data of the students, were used (Fig. 1). The APSQ is a 26-item questionnaire developed by Carruthers et al. in 2009 [15]]. It was adapted into Turkish and its validity and reliability study was conducted by Torun et al. in 2018, and two items were removed from the scale. It consists of 24 items and 9 sub-dimensions. It is a five-point Likert-type scale scored from 1 (strongly disagree) to 5 (strongly agree). In the evaluation of the scale, items 11, 13, 14, 15, 16, and 17 are reverse-coded. The maximum score to be obtained from the scale is 120. High scores indicate positive attitudes toward patient safety culture. Written permission was obtained from the authors via email to use the scale in the study [8]. In this study, the APSQ scale was selected due to its established Turkish validity and reliability in measuring patient safety perceptions. However, during the pilot application of the APSQ scale, some issues were identified. Certain questions were not correctly understood by the participants, and the utilization of a 7-point Likert scale in the initial version of the scale confused the respondents. Additionally, some statements did not load onto their respective factor dimensions as expected. Further analysis of the factor loadings revealed that the question "If people are more careful while doing their jobs, making medical mistakes can be avoided" exhibited a low value of 0.129 and was inverted in its loading. Even though it's inverted, the question "Patient safety problems cannot be taught only theoretically but can be learned through experience gained in the clinic" showed a high but negative factor loading. Considering these issues, a decision was made to remove two questions from the APSQ scale [8].The student information form consists of eight questions about the socio-demographic characteristics of the students.

In the second stage of the study, focus group interviews were conducted to reveal students' experiences and views on patient safety culture. The focus group interviews incorporated the use of the following semi-structured questions to guide the discussions and exploration of the students' perspectives:

1. Can you define patient safety?
2. What situations have you observed or experienced in your internship that put patient safety at risk?
3. What educational activities have you participated in so far that have raised your awareness about patient safety?
4. What kind of training should be conducted on patient safety in medical education, and what are your suggestions?

The interviews lasted about 45-60 minutes, with other questions to elicit more detailed information. The interviews were recorded, and the interview content was transcribed verbatim for content analysis.

2.1. Setting and sample

The participants were clinical experience semester students of a medical faculty. Those who volunteered to participate in the study between March 11 and May 11, 2022, were recruited for the study (4, 5, and 6year students) using the convenience sampling method.

2.1.1. Data collection

The data were collected in two stages. In the first stage, the "student information form," including sociodemographic information, and the APSQ created by the researchers were emailed to all clinical experience semester students in our faculty in the form of a Google survey. Quantitative data were collected from one hundred and twenty-five ($n = 125$) students. During the quantitative data collection phase, a total of 625 students were invited to participate in the study. Out of these, 125 students voluntarily agreed to take part, resulting in a participation rate of 20%. It is worth noting that the students who participated in the first phase of the study also gave their consent to be involved in the second phase.

In the second phase of the study, a focus group interview approach was employed. Students who received the questionnaire form were asked to participate in the focus group interview. Students who wanted to participate in the focus group interview were randomly selected and interviewed. Interview questions were formed in the context of the research hypothesis. The interviews were conducted by the researchers.

In the second stage of the study, voluntary students participated in focus group interviews conducted via Zoom. The interviews followed a semi-structured format, wherein students were encouraged to share their views and experiences related to patient safety. Qualitative data were collected through three separate focus group interviews, ensuring that data saturation was achieved, meaning that no new information or insights were emerging. A total of 18 students ($n = 18$) participated in these qualitative interviews.

2.2. Evaluation of the data

In the quantitative data analysis, descriptive statistics were employed to present continuous variables, expressed as mean ± standard deviation, while categorical variables were presented as n (%). For variables that exhibited a normal distribution, ANOVA (analysis of variance) and the Tukey test were utilized for conducting multiple group comparisons. Conversely, for those that did not follow a normal distribution, the Mann–Whitney U test was

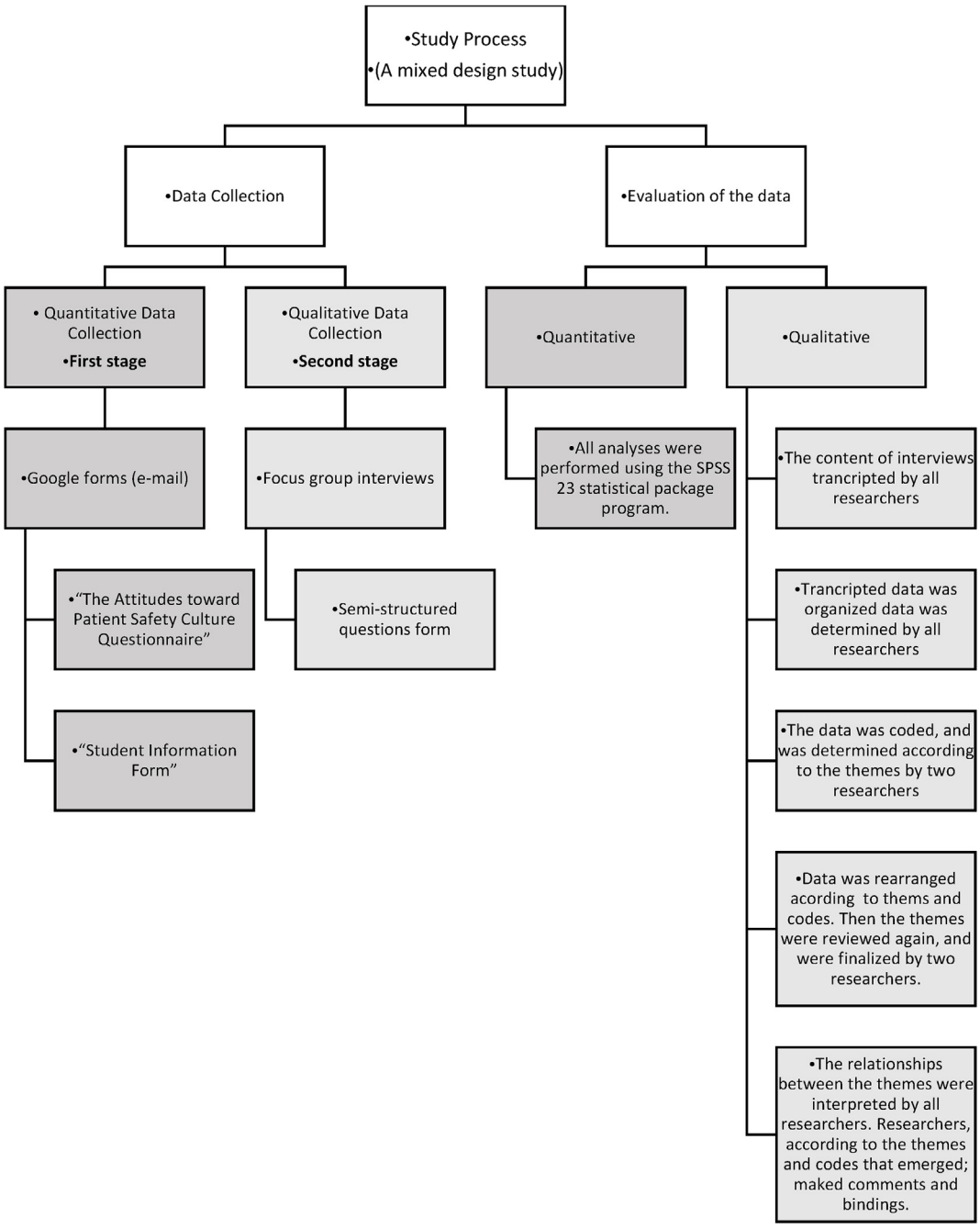


Fig. 1. Study design and data evaluation.

employed to assess differences between the two groups, and the Kruskal–Wallis test was utilized for multiple group comparisons. The $p < 0.05$ was considered statistically significant. The statistical analyses were conducted using the SPSS 23 statistical package program.

Content analysis is a methodological approach used to process and interpret data obtained from participants through interviews. It involves systematically coding the data to enhance its comprehensibility and organizing it into categories, thereby revealing underlying themes and relationships among these categories. The primary objective of content analysis is to uncover concepts and connections that can elucidate the collected data [16]. In our study, firstly the audio recordings were transcribed and converted into text. Secondly, the text was coded, and was created according to the subcategory by two researchers. The transcripts of the focus group interviews were subjected to content analysis separately by the researchers. The relationships between the themes were interpreted by all researchers. After that a consensus meeting was held on the themes by the researchers, and the themes were given their final shape (Fig. 1).

2.3. Ethics committee approval

Ethical approval for the study was obtained from the regional ethics committee with document number 24237859-162. Participants were informed about the purpose of the study, and their consent was obtained.

3. Results

3.1. A. Quantitative

Of the 125 students in the study, 68 (54.4%) were male. Fourth, fifth, and sixth-year students were 43 (34.4%), 42 (33.6%), and 40 (32.0%), respectively. The daily working time of 67 (53.6%) students in the hospital was 1-3 hours. Eighty-six (68.8%) did not receive training on patient safety. Of the students who received training on patient safety, 32 (82.1%) students received it from physicians during their internships. Sixty (48.0%) (Table 1).

The total score of the APSQ was 94.87 ± 14.32 (Table 2). A statistically significant difference was found between gender and the scores of the APSQ sub-dimension scores of working hours as error cause, patient's role in error, and error inevitability. Women stated that working hours are too long, errors are inevitable, and the patient plays a role in the development of medical errors. The APSQ scores did not show a difference between the descriptive characteristics of the students ($p > 0.05$). However, a statistically significant difference was found between the gender of the participants and the scores obtained from the APSQ scale's "working hours as error cause," "error inevitability," and "patient's role in error" sub-dimensions.

A statistically significant difference was found in the scores of the sub-dimension of working hours between genders ($p = 0.021$). A statistically significant difference was found in the scores of the sub-dimension of error inevitability between genders ($p = 0.021$). A statistically significant difference was found in the scores of the sub-dimension of the patient's role in error between genders ($p = 0.13$).

Working hours as the cause of medical errors, error inevitability, and the patient's role in the error were reported by women more than men (Table 3).

A statistically significant difference was found between knowing the definition of patient safety subscales and the scores of patient safety training received ($p = 0.002$). Comparisons between the groups showed that a difference arose between those who knew the definition of the concept of patient safety and those who did not. A statistically significant difference was also found between the status of knowing the definition of patient safety and the scores of the patient's roles in error ($p = 0.039$). This difference was caused by those who partially knew the definition of the concept of patient safety and those who did not know it at all (Table 3).

Table 1
Descriptive characteristics of the participants.

Socio-demographic characteristics		*n* (%)
Gender	Female Male	57 (45.6) 68 (54.4)
Marital status	Married Single	6 (4.8) 119 (95.2)
Year	4 year 5 year 6 year	43 (34.4) 42 (33.6) 40 (32)
Daily working hours	1-3 hours 4-6 hours >7 hours	67 (53.6) 33 (26.4) 25 (20)
Receiving training on patient safety	Yes No	39 (31.2) 86 (68.8)
The person providing patient safety training *	Physicians Nurse Online	32 (82.1) 5 (12.8) 2 (5.1)
Knowing the definition of the concept of patient safety	Yes Partially No	31 (24.8) 60 (48) 34 (27.2)
Age		Mean ± SD 23.92 ± 2.536

* The calculation was based on those who answered "Yes" to receiving patient safety training.

3.2. B. Qualitative

Six main themes emerged from the interviews: students' definitions of the concept of patient safety, students' misconceptions about "patient safety," students' knowledge about the scope of patient safety, situations witnessed by students that put patient safety at risk, deficiencies in educational processes on patient safety, and suggestions for educational approaches to patient safety. Sub-themes and details of the interview contents are given below.

3.3. Themes and interview contents related to sub-themes

3.3.1. Theme 1: Students' awareness of patient safety

3.3.1.1. Categories. Under the theme of students' definitions of the concept of "patient safety," the categories of physical safety of the patient and environmental safety of the patient emerged, and it was emphasized that some aspects of these categories need to be developed:

> *"…From a physical point of view, I think things like the presence of sharp instruments, an unsafe examination room, and a lack of personal space."*

> *"I think that the moment a person is admitted to the hospital as a patient is one of the moments, when s/he feels the most vulnerable, so the patient in this vulnerable and fragile state may be more open to external factors. So, the patient's environment can be organized accordingly, and I think s/he will feel safer in this way."*

3.3.2. Theme 2: Students' misconceptions about "patient safety"

3.3.2.1. Categories. Theme 2 was categorized as patient's mental safety and patient's material safety, and it was determined that students had deficiencies regarding the categories evaluated under the theme of misconceptions about "patient safety" and that they should receive training on patient safety:

> *"The psychological state of patients should be protected. For example, some patients may have mental problems, etc. Patients may have been discriminated against in some way, so they need to be examined in a safe environment without feeling that others think badly of them."*

> *"Patient treatment needs to be cost-effective. When I meet the patient, I should ask him /her "I plan your treatment in this way, can you afford it?" I mean, the patient should feel safe. If the treatment is very expensive, the patient has the right to know if s/he can afford it."*

Table 2
Scores obtained from the APSQ and its sub-dimensions.

Item		Total Mean ± SD
	The Attitudes toward Patient Safety Culture Questionnaire	94.87 ± 14.32
1	Patient safety training received	10.12 ± 2.70
2	Error reporting comfort	10.44 ± 2.80
3	Working hours as error cause	13.30 ± 2.70
4	Error inevitability	12.86 ± 2.64
5	Professional incompetence as error cause	12.28 ± 2.73
6	Disclosure responsibility	10.22 ± 2.78
7	Team functioning	8.86 ± 1.89
8	Patient's role in error	8.73 ± 1.81
9	Importance of patient safety in the curriculum	8.06 ± 2.08

Table 3
Comparison of the APSQ scale and sub-dimensions scores and descriptive characteristics.

Socio-demographic characteristics		n (%)	APSQ total score		Patient safety training received		Error reporting comfort		Working hours as error cause		Error inevitability		Professional incompetence as error cause		Disclosure responsibility		Team functioning		Patient's role in error		Importance of patient safety in the curriculum	
			X	SS	X	SS	X	SS	X	SS	X	SS	X	SS	X	SS	X	SS	X	SS	X	SS
Gender	Female	57 (45.6)	98.14	0.95	10.3	0.30	10.63	0.36	14.0	0.20	13.61	0.21	12.74	0.27	10.02	0.34	9.19	0.17	9.25	0.13	8.40	0.22
	Male	68 (54.4)	92.13	2.17	9.97	0.36	10.28	0.34	12.71	0.39	12.24	0.38	11.90	0.38	10.38	0.35	8.59	0.27	8.29	0.26	7.78	0.28
Test significance			U = 1.67		U = 1.86		U = 1.78		U = 1.50		U = 1.48		U = 1.72		U = 2.12		U = 1.76		U = 1.47		U = 1.71	
			p = .191		p = .729		p = .451		**p = .021**		**p = .021**		p = .284		p = .352		p = .321		**p = .013**		p = .236	
Year	Year IV	43 (34.4)	91.91	2.55	9.72	0.46	10.53	0.48	12.77	0.52	12.19	0.46	11.81	0.48	9.88	0.48	8.70	0.35	8.37	0.34	7.93	0.36
	Year V	42 (33.6)	96.98	1.95	10.60	0.36	10.55	0.39	13.43	0.32	13.17	0.35	12.40	0.39	10.69	0.37	8.93	0.25	9.12	0.19	8.10	0.29
	Year VI	40 (32)	95.85	2.03	10.05	0.41	10.23	0.42	13.73	0.36	13.28	0.38	12.65	0.37	10.08	0.42	8.98	0.24	8.70	0.27	8.18	0.30
Test significance			X^2_{KW} = 2.51		X^2_{KW} = 1.50		X^2_{KW} = 0.30		X^2_{KW} = 1.64		X^2_{KW} = 3.54		X^2_{KW} = 1.14		F = 0.96		X^2_{KW} = 0.87		X^2_{KW} = 2.16		X^2_{KW} = 0.03	
			p = .285		p = .471		p = .860		p = .440		p = .170		p = .565		p = .383		p = .958		p = .339		p = .982	
Daily working hours	1-3 hours	67 (53.6)	93.78	1.77	10.10	0.31	10.45	0.34	13.09	0.35	12.63	0.33	12.10	0.32	10.00	0.36	8.84	0.24	8.69	0.23	7.88	0.27
	4-6 hours	33 (26.4)	94.79	3.10	10.30	0.52	10.55	0.50	12.91	0.51	12.55	0.52	12.21	0.56	10.85	0.40	8.55	0.37	8.58	0.34	8.30	0.35
	>7 hours	25 (20)	97.92	1.32	9.92	0.53	10.28	0.53	14.36	0.20	13.92	0.29	12.84	0.44	9.96	0.57	9.36	0.19	9.04	0.22	8.24	0.31
Test significance			X^2_{KW} = 1.85		X^2_{KW} = 0.59		X^2_{KW} = 0.48		X^2_{KW} = 5.57		X^2_{KW} = 4.46		X^2_{KW} = 1.42		F = 1.16		X^2_{KW} = 1.65		X^2_{KW} = 0.21		X^2_{KW} = 0.93	
			p = .395		p = .743		p = .783		p = .062		p = .107		p = .489		p = .316		p = .438		p = .898		p = .627	
Knowing the definition of patient safety	Yes	31 (24.8)	93.45	3.71	10.87	0.60	10.68	0.61	12.58	0.63	12.03	0.65	12.16	0.61	9.74	0.53	8.71	0.41	8.58	0.42	8.10	0.42
	Partially	60 (48)	94.82	1.44	10.20	0.30	10.47	0.30	13.30	0.30	12.93	0.28	12.20	0.30	10.52	0.34	8.96	0.78	8.57	0.20	7.83	0.25
	No	34 (27.2)	96.26	2.12	9.29	0.41	10.18	0.50	13.94	0.40	13.50	0.37	12.53	0.47	10.12	0.47	9.12	0.31	9.15	0.28	8.44	0.34
Test significance			=2.47		X^2_{KW} = 12.04		X^2_{KW} = 1.51		X^2_{KW} = 5.22		X^2_{KW} = 3.76		X^2_{KW} = 1.06		F = 0.81		X^2_{KW} = 2.23		X^2_{KW} = 6.51		X^2_{KW} = 3.16	
			p = .291		**p = .002**		p = .469		p = .074		p = .152		p = .586		p = .444		p = .327		**p = .039**		p = .205	

(X^2_{KW}= Kruskal–Wallis test, X = Arithmetic mean, SD= Standard deviation, U = Mann-Whitney U test).

3.3.3. Theme 3: Students' knowledge about the scope of patient safety

3.3.3.1. Categories. This theme consisted of the following categories: infections transmitted during healthcare delivery, verification of patient credentials, risk of falls, medication administration, and complications during healthcare delivery. Students expressed their opinions on the categories of this theme in line with the experiences they gained during their internship practices. In this context, it is noteworthy that students were not aware that nontechnical skills like communication would also put patient safety at risk.

"*There was a situation that I understood much better during the COVID pandemic. We were entering the patient room as a crowded group during the visit, and there was a possibility of infecting the patient with COVID.*"

"When we talk about patient safety, first, the patient must be correctly identified. Sometimes there can be confusion during drug administration or blood sampling."

"*Patients may have a risk of falling due to their illness. We stick a warning on their doors that there* may *be a risk of falling.*"

"I think the prevention of complications that may occur after the treatment or after the procedure is included in patient safety. Complications that may occur before, during, and after diagnosis, and treatment should be prevented."

3.3.4. Theme 4: Students' experiences with medical errors

3.3.4.1. Categories. The categories are listed as situations related to patients, patient relatives, diagnosis and treatment processes, health professionals, and organizational problems. The negative experiences of the students can be scripted and integrated into the patient safety training module.

"*Sometimes when interventions are needed, the relatives of the patients want to witness the interventions, they want to be there. If something develops at that moment, there may be a complication, or even just the presence of the patient's relatives at that moment can sometimes be a situation that jeopardizes the safety of the patient for the health personnel performing the procedure.*"

"I witnessed the physician ordering the wrong test at a time when the outpatient clinic was very busy. When we talked about it later, he stated that he had asked for the wrong test several times before due to the heavy workload of the clinic."

"Sometimes in the wards, for example, there are two patients with the same name, blood is taken without paying attention to their surnames and sent to the laboratory. Of course, the patient's results are wrong. Therefore, they take blood from both the patient they should *normally take and the other patient again. Also, when a procedure is to be performed, there may be confusion about the name. For example, a patient is told that they will have an ultrasound. Later, it is understood that an ultrasound will be performed on another patient, not on that patient. When the patient is told that an ultrasound will not be performed, the patient's confidence is shaken, and they can be afraid."*

"*There is a lack of materials in some wards. Sometimes there are no sterile gloves.*"

3.3.5. Theme 5: Deficiencies in educational processes on patient safety

3.3.5.1. Categories. Failure to create a suitable training environment in clinics and inadequacies in practices also indicate a lack of training on patient safety. In addition, the fact that students are

expected to practice in clinics where they do not feel competent puts patient safety at risk.

"*While studying other subjects, we received partial training on patient safety. In practical training, we were taught to ensure our safety first and then the patient's safety. However, I think that this training did not create awareness in us until we started the internship. During the internship, we learned by experience, by seeing, by doing.*"

"*During my internship in the internal medicine service, they asked me to insert a urinary catheter in the tenth minute of my first shift. In the 4th year, we practiced. In the 5th year, we received distance education due to the pandemic. I had no experience real patient inserting a urinary catheter. As a result, such a situation can be a problem for the patient's health.*"

3.3.6. Theme 6: Suggestions for educational approaches to patient safety

3.3.6.1. Categories. The categories under this theme are simulation-based training, on-the-job training, time of training, and training delivery. Regarding the categories of this theme, students made suggestions for receiving structured patient safety training.

"*Training on patient safety can be organized using various scenarios in simulated environments. Cases can be given, and patient safety can be discussed in them.*"

"Training on patient safety should be given in the field. The examination time we allocate to patients should be increased so that we can make the right diagnosis and provide the right treatment. In addition, we can do internships not only in university hospitals but also in community healthcare centers or state hospitals."

"There should be training starting in the second half of semester 3 so that we are ready for the clinical period. We take a module course on personnel safety, and patient safety can be added here. Short module training can be given before starting an internship to remind students about patient safety."

"Patient safety education should be presented as a whole; it does not attract our attention when it is given within other practices. A focused and concentrated education should be given."

4. Discussion

We interpreted the results of the quantitative and qualitative analyses, respectively.

Patient safety is a significant issue in healthcare provision, so prospective physicians should graduate with an awareness of this issue. In our study, the score of 94.87 $\pm$ 14.32 obtained from the APSQ scale of patient safety culture is considered positive. In contrast to our study, several other research endeavors focusing on medical students and patient safety have shown that there is inadequate development of patient safety culture and perceptions among this population [17,18]. These studies have reported that errors in patient care have been attributed to factors such as ignorance, miscommunication, and issues related to medication administration [19]. Although the students did not receive training on patient safety, the high score obtained from the patient safety attitude scale is thought to be due to the culture they acquired from the institution where they work. In this case, it is thought that the hidden curriculum is effective.

The APSQ scores did not show the difference between the descriptive characteristics of the students [20]. Nonetheless, it was observed that female students achieved higher scores in specific sub-dimensions of the APSQ scale, including "working hours as error cause," "error inevitability," and "patient's role in error." Similar findings were reported in a previous study [20]. Conversely, in a study focusing on patient safety, male students exhibited more positive attitudes compared to female students. In that study, female students tended to associate errors with factors such as working hours and professional incompetence to a greater extent [17]. These differences in attitudes may be linked to the higher sense of responsibility that female students often perceive.

More than one-third of the students did not receive any training on patient safety. In our study, a majority of the students indicated that they possessed a partial or comprehensive knowledge of the definition of patient safety. This contrasts with a study that specifically examined nursing and medical students' patient safety knowledge and found that the students' overall knowledge level in this area was deemed insufficient [18]. Students who knew the definition of patient safety reported that the patient had a role in the error. The reason for this is thought to be related to the experiences of the students.

Interestingly, a considerable number of students who demonstrated an understanding of the patient safety definition also recognized the patient's role in errors. This noteworthy observation could potentially be linked to the students' personal experiences. It is plausible that their exposure to real-life situations or clinical practice has contributed to a heightened awareness of the patient's involvement in errors. Consequently, this increased awareness might have influenced their perceptions and responses concerning patient safety concepts.

4.1. Students' awareness of patient safety and students' misconceptions about "patient safety"

The focus group interviews revealed that students expressed a lack of structured training on patient safety culture. This suggests that there might be misconceptions among students regarding the definition of patient safety culture. However, in our study, students defined patient safety as encompassing physical, mental, material, and environmental safety. Moreover, they highlighted the significance of patients being informed about the cost of treatment before the commencement of healthcare services. Students perceived this situation as jeopardizing patient safety, as it could impact patients' decision-making regarding treatment acceptance or refusal and potentially affect their mental well-being. Additionally, financial constraints limiting access to healthcare services were also identified as factors potentially compromising patient safety. However, these aspects might not solely relate to patient safety but rather fall under the domain of health policies and health advocacy. According to the definition of the Institute of Medicine (IOM), patient safety is used in a broad sense and is defined as preventing harm to the patient. The American National Patient Safety Foundation (ANPSF), on the other hand, defined it as "preventing healthcare-related errors and reducing patient harm caused by these errors" [21]. However, in our study, students defined patient safety as physical, mental, material, and environmental safety. It is interesting to note that the students involved in our study considered the fact that patients are not informed about the cost of treatment before the start of health service provision as a situation that jeopardizes patient safety in the acceptance or refusal of treatment, that this situation would impair the mental health of the patients, and that financial constraints restricting receiving health services would jeopardize patient safety. As a result rather than being a topic related to patient safety, it is a situation that concerns health policies and health advocacy. Students have misconceptions about patient safety and should receive curriculum-integrated training on patient safety. Patient safety is a significant issue in healthcare

provision, so prospective physicians should graduate with an awareness of this issue.

4.2. Students' knowledge about the scope of patient safety

In our study, students' definition of patient safety primarily focused on specific aspects such as patient identity verification, infections, the risk of falls, medication administration, and complications related to healthcare utilization. Surprisingly, they did not mention other critical dimensions of patient safety, including aspects related to teamwork, communication between healthcare professionals, and leadership. It is important to note that nontechnical skills, such as effective communication, play a pivotal role in preventing errors and ensuring patient safety [22]. In a study, students stated that they could understand very little about patient safety issues without participating in a module [23]. It is necessary to structure programs that use active learning strategies in simulated environments that will teach our students nontechnical skills as well as the theoretical issues of patient safety.

4.3. Students' experiences with medical errors

In the focus group interviews, students identified several factors that they perceived as putting patient safety at risk. These factors encompassed various aspects of health care, including issues related to patient relatives, the diagnosis and treatment processes, interactions between the patient and healthcare professionals, as well as organizational problems within healthcare settings. Notably, the comments made by the students during the interviews were largely derived from their clinical experiences. The report on the causes and prevention of medical errors highlighted several factors leading to such incidents, including errors during surgery, diagnosis, medication administration, malfunctioning of devices and equipment, infections, falls, and issues related to healthcare technology [24]. As mentioned earlier, students' evaluations of medical errors are often influenced by their personal experiences in clinical settings [24]. Indeed, medical students' experiences during their clinical rotations can play a pivotal role in shaping their attitudes toward patient safety and subsequent behaviors as future healthcare practitioners. A study showed that unprofessional behavior in physicians may be associated with poor care collaboration, patient dissatisfaction, and increased medical errors [25].

4.4. Deficiencies in educational processes on patient safety

The findings of our study revealed that students expressed a lack of training on patient safety and identified deficiencies in the practice of their training within clinical settings. These inadequacies included instances where students were expected to perform procedures regardless of their level of competence and instructors' inability to allocate sufficient time for students in clinical practices, both of which were perceived to jeopardize patient safety. A previous study focusing on students and patient safety echoed similar sentiments, wherein students expressed a desire to undertake clinical practices with trainers and felt uncomfortable and unsupported when trainers were not present [23]. In this sense, trainers need to support students first in simulated environments and then in on-the-job applications and provide feedback to them.

4.5. Suggestions for educational approaches to patient safety

The insights from our study highlighted students' suggestions concerning the method and timing of patient safety education. They believe that providing patient safety education in simulated environments during situations that jeopardize patient safety would enhance its memorability and practical application. Furthermore, students emphasized the significance of on-the-job training under the guidance and supervision of trainers as a valuable means to ensure patient safety. Considering the structure of medical education in our institution, where the first three years are dedicated to theoretical courses, and the last three years involve clinical experiences with patients; students stressed that patient safety training should be delivered before their interactions with patients in the initial three years. Moreover, they recommended providing reminder training on patient safety before commencing their internship in the 6th year. In a study, Thyness and colleagues found that practical training positively influences students' self-confidence [23]. In another study, trainers showed motivation in developing students' teamwork, leadership, and communication skills, thereby contributing to an improved patient safety culture [24]. In our study, students expressed that they would like to receive training on patient safety.

5. Conclusion

In our study, we found that students demonstrated a positive attitude toward patient safety culture; however, they exhibited a lack of knowledge concerning patient safety issues. This positive attitude is believed to be influenced by their experiences in clinical environments. Structured training based on simulation can be given to students to improve their knowledge of patient safety and improve their attitudes and behaviors.

6. Limitations

Our study assessed the attitudes and opinions of students regarding patient safety in a medical faculty. It is essential to acknowledge certain limitations. Conducting survey-based studies to measure behaviors, such as attitudes, in students has inherent limitations. To gain a deeper and more comprehensive understanding of student perspectives, longer-term studies using qualitative research designs with ethnographic observations could be considered in the future.

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CRediT authorship contribution statement

Yasemin Güner: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kübra Yilmaz:** Data curation, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Dilan Yurtseven:** Data curation, Formal analysis, Investigation, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Tuğçe Mümin:** Data curation, Resources, Software, Writing – original draft, Writing – review & editing.

Conflicts of interest

As corresponding author I have authority over manuscript preparation and decisions to submit the manuscript for publication. Besides I am aware that any of the authors' academic institutions have not any financial interest in or a financial conflict with the subject matter or materials discussed in this manuscript. In

addition to our manuscript is not published previously, and will not be submitted for publication elsewhere.

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